

Part C — AI Usage Report

DSC 190/291 — Assignment 6 Student: Zeyu Bian

1. Where I used AI

- **A.1 Hölder step.** I wrote the margin-inequality-take-expectation step myself, then asked AI to double-check that ℓ_1/ℓ_∞ Hölder is exactly what's needed (and not ℓ_2/ℓ_2). AI confirmed and helped me write the bound in the cleanest form via $|\mathbb{E}_Q[y\varphi_j]|$ rather than the signed quantity.
- **A.3 hard construction.** This is the most AI-assisted part. The handout's hint describes three column families ("base / pair-flip / carry") and the weighted-sum constraint, but I wanted to be sure the carry-column construction actually gives q -weighted sum 1 for *every* p . I wrote out the cases by hand on paper, then asked AI to verify the algebra. We caught one off-by-one in my initial carry column for $p = 1$.
- **B.1 case split.** I had the "constant halfspace for small p , averaging for large p " outline from the hint. AI helped me find the right form of the inequality $\sum_i q_i \leq (k-1)(1-p)$ via union bound on $\{h_i(x) = -1\}$ events, and then algebraically reduce $(k-1)(1-p)/k$ to $\leq 1/2 - 1/(2k^2)$ for the boundary case.
- **A.4 experiment.** I wrote the AdaBoost loop and the data generators. AI suggested the right diagnostics to log (edge per round, support size, ℓ_1 -normalized margin), and helped track down a bug where I was computing the edge under the *uniform* sample distribution instead of the current D .

2. AI suggestions I accepted (and why)

- **"Take expectation of the margin inequality" framing for A.1.** AI suggested writing the proof as: (i) margin $\Rightarrow \mathbb{E}_Q$ of the inner product ≥ 1 , (ii) Hölder. I accepted because this makes the $1 \leq \|w^*\|_1 \cdot \max_j |\mathbb{E}_Q[y\varphi_j]|$ identity a *one-line* derivation rather than a multi-step manipulation.
- **Edge formulation** $L_D(b) = (1 - \sum_i D_i y_i b(x_i))/2$. AI proposed deriving the $O(nd)$ weak learner by reducing weighted error to a signed correlation, so the algorithm is "compute $c_j = \sum_i D_i y_i \varphi_j(x_i)$ for each j , return $b_{\arg \max_{|c_j|, \text{sign}(c_j)}}$." This is the cleanest expression of the weak learner.
- **$\|w^*\|_1 \geq Z$ argument in A.3.** Instead of trying to compute $\Phi^{-1}\mathbf{1}$ in closed form, AI suggested using the A.1 contrapositive: best coordinate edge $\leq 1/\|w^*\|_1$ and best coordinate edge equals $1/Z$, so $\|w^*\|_1 \geq Z$. Then $\|w^*\|_\infty \geq \|w^*\|_1/s = 2^{\Omega(s)}$. Much cleaner than a direct matrix-inverse computation.
- **B.1 boundary algebra.** AI suggested writing $(k-1)(1-p)/k \leq 1/2 - 1/(2k) + 1/(2k^2) - 1/(2k^3)$ and using $k \geq 2$ for the final step. I accepted; it makes the inequality verifiable line-by-line.
- **B.3 sample bound $\tilde{O}(k^4 d/\varepsilon)$.** AI walked through the $T = O(k^4 \log n)$ rounds $\times$ VC dim $\tilde{O}(Td) \times$ realizable rate $1/\varepsilon$ composition. I verified each step.

3. AI suggestions I rejected or substantially modified

- **A.3: AI's first carry-column proposal.** AI initially gave a carry column with $\Phi_{(r,+),(p,-)} = +1$ for $r < p$ (which would give $\sum_{r < p} q_r \cdot 0 = 0$ instead of $-(2^p - 2)$). I caught it by computing the weighted sum for $p = 2$: the proposed column gave a sum of $-1 + 0 + 2^2 = 3 \neq 1$. I rewrote the carry to use $(-, -)$ for $r < p$.

- **A.3: AI tried to invoke a Hadamard-style lower bound.** AI suggested skipping the explicit construction and quoting “any ± 1 matrix with all column-sums equal to 1 under exponential weights has a $2^{\Omega(s)}$ inverse.” I rejected because the construction is part of the problem; the existence claim has to be made concrete.
- **B.1: AI’s first attempt used $L_{\mathcal{D}}(h_i) \leq (1 - p)$ directly.** This is a trivial bound that doesn’t beat $1/2$ when $p < 1/2$. I corrected by introducing the union-bound step $\sum_i \Pr[y = -1, h_i = -1] \geq \Pr[y = -1]$ to get the factor $(k - 1)/k$.
- **B.2: AI wanted $\varepsilon_0 = 1/(2k^2)$.** This would give the weak learner edge 0 in the boundary case. I changed to $\varepsilon_0 = 1/(4k^2)$, giving edge $\gamma = 1/(4k^2) > 0$. The factor of 4 matters: B.3’s round count becomes $T = O(k^4 \log n)$ instead of “undefined.”
- **B.4: AI mixed up uSVP and RSAT regimes.** AI’s first draft applied uSVP to $k = \omega(1)$ and RSAT to $k = d^r$. I corrected: uSVP gives hardness for polynomially-large $k = d^r$; RSAT gives hardness for any $\omega(1)$ size including sub-polynomial.

4. How I verified correctness

- **A.1.** Verified Hölder by considering the extremal case where all mass of w^* concentrates on one coordinate ($w^* = Be_{j^*}$): then the coordinate predictor on that coordinate has edge $\geq 1/(\|w^*\|_1) = 1/B$, matching the bound.
- **A.2 round count.** Verified $T = O(s^2 B^2 \log n)$ matches Schapire–Singer’s $T = O(\log n / \gamma^2)$ with $\gamma = 1/(2sB)$.
- **A.3 column-sum identity.** Verified in `experiment.py` line `assert np.allclose(sums, 1.0)` — the 7×7 matrix actually does satisfy $\Phi^\top q = \mathbf{1}$ as constructed.
- **A.3 $\|w^*\|_\infty$ size.** Computed by `np.linalg.solve(Phi, np.ones(s))` and confirmed $\|w^*\|_\infty = 9 = 2^m + 1$, comfortably $2^{\Omega(s)}$ for $m = 3$, matching the lower bound from A.1.
- **A.4.** Verified the AdaBoost loop on the easy distribution against a one-line sanity check: training error reaches 0 in $O(\log n / \gamma^2)$ rounds, support matches the planted support, ℓ_1 -margin is close to the optimum $1/\|w^*\|_1$.
- **B.1.** Verified the case split numerically for $k = 2, 3, 4$: I computed $(k - 1)/k \cdot (1/2 + 1/(2k^2))$ explicitly and confirmed it’s $\leq 1/2 - 1/(2k^2)$ in every case.
- **B.2/B.3 polynomial-time bookkeeping.** Wrote down the chain: $\varepsilon_0 = 1/(4k^2) \Rightarrow \gamma = 1/(4k^2) \Rightarrow T = 8k^4 \ln n \Rightarrow \text{VCdim} = \tilde{O}(Td) = \tilde{O}(k^4 d) \Rightarrow n = \tilde{O}(k^4 d / \varepsilon)$. All polynomial in d when $k \leq d^c$.
- **B.4 contradiction direction.** Verified by contrapositive: hardness fact + implication $\Rightarrow$ negation of “halfspaces are agnostically PAC learnable.”

AI workflow updates (5 most recent changes)

1. **“Compute the constants in case-split inequalities” rule.** Both A.3 (carry-column sum) and B.1 (the $(k - 1)(1 - p)/k \leq 1/2 - 1/(2k^2)$ reduction) had off-by-one / sign-flip bugs from AI on the first try. I now require that any case-split inequality in a proof comes with an explicit numerical check at the boundary value ($p = 1/2 - 1/(2k^2)$ here) before I write it down.
2. **“Use the contrapositive of part 1 rather than reproving” rule for A.3.** Rather than computing $\Phi^{-1}\mathbf{1}$ directly to bound $\|w^*\|_\infty$, the cleaner argument is: A.1 says the best edge is $\leq 1/\|w^*\|_1$, and we constructed Φ so the best edge is $1/Z$, so $\|w^*\|_1 \geq Z$. I added a note to my workflow: “for tightness results, see if the upper-bound theorem can be contrapositive’d into a matching lower bound.”
3. **“AdaBoost diagnostics” checklist.** Whenever I implement boosting, I now log a fixed five

metrics per round: train 0-1 error, exponential loss, observed edge, support size, ℓ_1 -normalized margin. This caught the “edge computed under uniform D ” bug in A.4: my first run had edge ≈ 0.4 in every round, which is impossible on the hard distribution with $1/Z \approx 0.067$.

4. **“Distinguish weak / strong PAC learnability assumptions” prompt.** B.2/B.3 ask for the agnostic learner $\rightarrow$ realizable learner direction. I now ask AI explicitly to state which type of PAC learner is hypothesized and which is concluded, to avoid the easy slip “if X is realizable then Y is agnostic” (the directions matter).
5. **“Get the hardness regime right” check.** I now read out loud “uSVP gives X -size hardness; RSAT gives Y -size hardness” before applying either, since AI mixed them up in B.4’s first draft. Reading conjectures out loud is a low-tech but effective defense against AI’s overconfidence on cryptographic-assumption names.

I did not use AI to fabricate any reference; the standard AdaBoost training-error analysis (Schapire–Freund), Hölder, and the boosted-class VC bound (Schapire et al. 1998) are all results I have read.